

A portrait of Afonso Caboz, a man with short dark hair and a light beard, looking slightly to the left. He is wearing a grey shirt. In the background, there are several flags, including the European Union flag with its yellow stars on a blue field.

AFONSO CABOZ

Full-Stack Solutions Architect

afonso.caboz@gmail.com | linkedin.com/in/afonsocaboz | github.com/CodeZobac | codezobac.com | Warsaw, Poland

My development philosophy is built on a dual-core engine of reflection and expansion. I begin with the **Yin** phase: a period of deep research and empathetic UX mapping, where I inhabit the user's perspective to ensure every feature is necessary and intuitive. Once the architecture is solidified, I transition into the **Yang** phase: an aggressive, AI-augmented execution cycle focused on implementation, automated testing, and full-lifecycle ownership. Rather than getting lost in the minutiae of syntax, I act as a technical architect—directing AI agents with precise intent and verifying success through the only metric that matters: a seamless, functional, and high-performance user experience.

VIVADRIVE

My tenure at VivaDrive was defined by a singular mission: resolving high-stakes engineering bottlenecks through architectural transmutation. I did not merely fix errors; I re-engineered systems to turn volatility into stability. This was achieved through four critical pillars of innovation:

I. Cost Engineering & LLM-as-a-Judge

When faced with the economic friction of API budget saturation, I re-architected our OpenAI integration. By implementing a strategic "LLM-as-a-Judge" framework using `gpt-4o-mini`, I transmuted a manual, expensive evaluation process into an automated, programmatic pipeline. This allowed us to achieve radical cost reduction while simultaneously elevating the precision of our output quality.

II. Illuminating the Black Box: Traceability & Observability

To resolve the opacity inherent in complex LLM workflows, I engineered a four-layer observability framework utilizing **Langfuse**. This architecture provided total visibility across the entire request lifecycle—tracing every movement from the "Root" (monitoring API expenditure and authorship) to the final "Generation." This brought much-needed transparency and accountability to our generative processes.

III. From Static Data to Predictive Intelligence

I led the evolution of our data architecture from reactive, static file processing to a proactive **"Fuel Intelligence Layer."** By transitioning toward a predictive Data Warehouse model, I enabled the system to move beyond simple storage, empowering it to detect anomalies and identify fuel fraud patterns through advanced algorithmic modeling.

IV. Managing Autonomy: The Self-Correcting Ecosystem

The most complex challenge was managing the inherent entropy of autonomous agents. As I worked side by side with the tech lead to drive the macro-refactoring of our platform from simple assistants to fully autonomous entities, I encountered the risk of model hallucinations. To neutralize this, we engineered an **"AI Automated Testing"** pipeline—a self-correcting production environment where specialized models are tasked with validating the outputs of others, ensuring that autonomy never comes at the expense of truth.

ETIC ALGARVE

Prior to my work with distributed AI systems, my tenure at ETIC Algarve provided a profound lesson in the necessity of structural integrity when confronting systemic decay. I was tasked with managing a legacy resource management application that had reached a state of functional obsolescence—a system burdened by technical debt and incapable of scaling with institutional needs.

I. Architectural Audit & The Rejection of Superficiality

Upon conducting a deep-dive technical audit, I identified that the existing architecture was suffering from terminal entropy; the lack of responsiveness was not a surface-level UI issue, but a fundamental failure in the underlying structure. While a superficial "patch" could have provided temporary relief, I made the strategic decision to advocate for a complete architectural rebuild—a move designed to ensure long presence and long-term maintainability.

II. Rapid Manifestation via Prototyping

To validate this high-stakes pivot, I entered an intensive development cycle, engineering and presenting a functional prototype within just seventy-two hours. This phase required a high-velocity execution transforming abstract architectural concepts into a tangible, demonstrable reality to prove the viability of the rebuild.

III. The Synthesis of Technical Truth and Stakeholder Alignment

This period served as my most critical lesson in professional engineering: that technical excellence is hollow without organizational resonance. I learned that for an architectural shift to succeed, the phase of stakeholder alignment, navigating feedback loops, managing institutional expectations, and refining communication protocols is just as vital as the code itself. True innovation requires that technical truth be synchronized with organizational objectives.

IV. MVP Delivery & Empirical Validation

With the institution's mandate secured, I transitioned into a full-scale development sprint, delivering a production-ready MVP within one week. The project culminated in a high-stakes defense before an assembly of industry professionals and educators, where the architecture was validated not merely by its code, but by its ability to resolve complex, real-world operational requirements with precision.

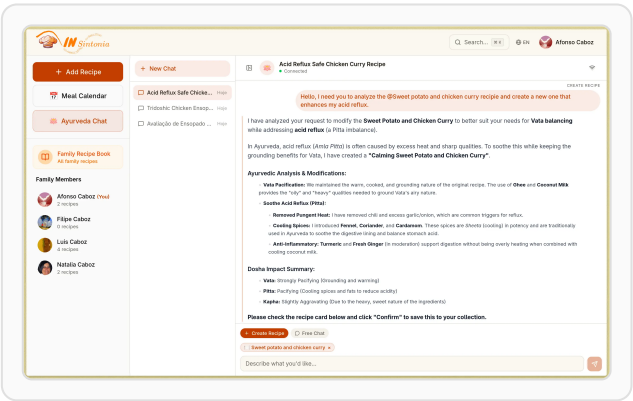
Selected Works

I. In Sintonia

Ayurvedic Nutritional Intelligence

An agentic real-time platform engineered to align dietary interventions with Ayurvedic principles through high-precision orchestration.

- **Minimalist Agentic Orchestration:** Engineered a highly efficient, **Mixture-of-Experts (MoE)** architecture using a hierarchical system of prompts. Rather than managing the complexity of the final output, I focused on the precision of the input—implementing a "root-level" verification layer that ensures each sub-agent receives only the context necessary for deterministic relevance, effectively eliminating semantic drift.
- **High-Performance Rust Infrastructure:** Architected a robust backend using **Rust** and **ZeroClaw** for low-latency AI orchestration. The system utilizes **Axum + Utopia** for high-throughput API performance, integrated with **WebSockets** to facilitate seamless, real-time nutritional guidance and continuous data streams.

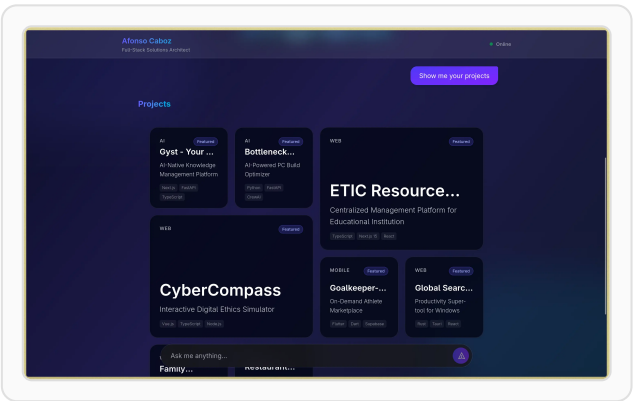


II. Personal Chatbot

Generative UI & Dynamic Knowledge Interface

An experimental implementation of a generative interface that transforms natural language queries into structured, component-based visual experiences.

- **Dynamic Component Manifestation:** Engineered a "Generative UI" paradigm where the interface is not static but reactive. The system interprets user intent through an AI-driven layer, triggering the on-demand rendering of specific UI components (experience timelines, skill matrices, project deep-dives) based on the semantic requirements of the query.
- **Intelligent Knowledge Orchestration:** Integrated a dynamic, retrieval-augmented knowledge base that allows the agent to navigate my professional history with precision. This architecture ensures that every interaction—whether answering technical inquiries or presenting biographical data—is grounded in an authoritative and contextually accurate data layer.



Ultimately, my trajectory in software engineering has been defined by a single, continuous cycle: the ability to identify structural entropy and transmute it into scalable, high-performance architecture. Whether I am auditing legacy systems to prevent technical decay or orchestrating complex, AI-driven ecosystems to drive unprecedented efficiency, my objective remains constant: **to bridge the gap between profound technological complexity and intuitive, user-centric reality.**

I do not merely write code; I engineer environments where technology serves a clear, strategic purpose. My expertise lies in navigating the delicate tension between the **Yin** of deep, architectural research and the **Yang** of rapid, high-impact deployment. I bring to your team a commitment to technical excellence, experience with reducing operational friction, and a proven methodology for transforming ambitious digital visions into resilient, production-ready realities.

I am prepared to apply this architectural rigor to your most complex challenges, ensuring that every system I touch is built not just to function, but to endure and evolve.